Use work-energy to solve motion

antocciphysics | TEACHER KEY | CPA + MATCHING SEI

Learning target: Choose work-energy when force and displacement are known.

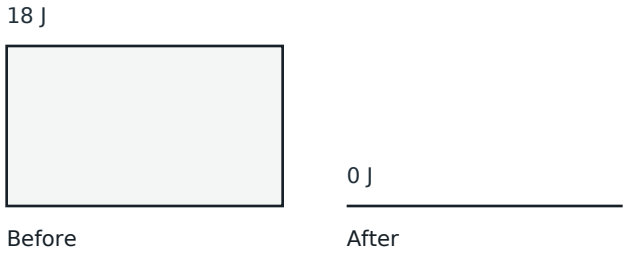
Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 4, Day 2.*
+24 J; force and displacement point the same way.

02. Worked example: explain the model
Net force along motion; energy grows from 0 to 12 J. Work-energy uses displacement without needing elapsed time.

03. Read and interpret the model
-18 J; final kinetic energy 0.

Model / diagram check



04. Apply the physics
4.5 m.

05. Explain your reasoning
Object likely stopped before assumed endpoint; braking model interval cannot continue with negative kinetic energy.

06. Review: connect an earlier idea *Review: Unit 3, Day 7.*
East positive: 6-4=2 kg·m/s east. Opposite directions require opposite signs.

07. Exit check
20 J.

08. STRETCH: test the model
Distances 6 and 3 m; larger opposing force stops it in less distance.

Use work-energy to solve motion

antocciphysics | TEACHER KEY | HONORS + MATCHING SEI

Learning target: Choose work-energy when force and displacement are known.

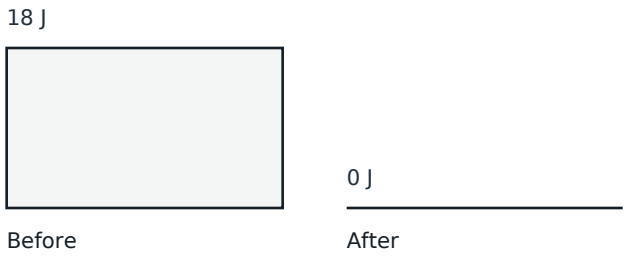
Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 4, Day 2.*
F=5 N; zero displacement gives 0 J.

02. Worked example: explain the model
Net force along motion; energy grows from 0 to 12 J. Work-energy uses displacement without needing elapsed time.

03. Read and interpret the model
-18 J; final kinetic energy 0.

Model / diagram check



04. Apply the physics
Initial 36 J; distance 12 m. Double speed gives 4 times energy/distance: 48 m.

05. Explain your reasoning
Object likely stopped before assumed endpoint; braking model interval cannot continue with negative kinetic energy.

06. Review: connect an earlier idea *Review: Unit 3, Day 7.*
Deviation≈5%. It compares values relative to the initial value; uncertainty requires a measurement/error analysis.

07. Exit check
Net 40 J; v=√(80/5)=4 m/s.

08. STRETCH: test the model
Change in K=2-50=-48 J; force=-48/6=-8 N.